

**CatSight AI: Enhancing Document Search Accuracy and User Interaction through OCR,
Vector Similarity Search, and LLM-Based Chat Interfaces**

A Thesis Project

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Revision Summary

Revision No.	Details	Section / Page No.
1	Missing Theoretical Framework	Methodology, Page 34
2	Be proper and consistent with the figure sizes, and make it clearer	Methodology Page 40
3	Add Design and Implementation Issue	Methodology, Page 34
4	Proper spacing	Methodology. Pag 44
5	Missing Summarization Figure flow	Methodology, Page 44
6	Use Rouge, Bleu, Bertscore for Summarization Metrics	Methodology
7	Mention the figures that are shown	Methodology
8	Use LaTeX Equation	Methodology
10	Provide cosine similarity score equations	
11	Explain Token and Cosine Similarity Score	
12	Discuss the map reduce in the paper	
13	Write algorithm for generate summary	
14	Define hallucination in LLM and its basis	
15	Explain zero-shot method	
16	Specific objective be more concise	Specific Objective Page 4
17	Show a architecture diagram to show the interactions of each component	3.2 methodology. Page 37
18	Express the diagrams in algorithm instead of flow charts	

Abstract

The CatSight AI system addresses the limitations of MSU-IIT’s keyword-based document search by leveraging a unified pipeline of OCR, vector embeddings, and an LLM-driven conversational interface. The architecture follows a retrieval-augmented generation (RAG) framework: documents (scanned or digital) are processed via OCR (ensuring text extraction from scanned or image-based documents), segmented into chunks, embedded into vector representations stored in a vector database, and then retrieved to support LLM response generation. This integration enables users to issue intuitive natural-language queries and receive contextually relevant answers through a chat-based interface.

In evaluations against the legacy system, CatSight AI achieved higher retrieval accuracy, semantic relevance, and efficiency. The hybrid approach substantially improves semantic understanding, capturing paraphrases and synonyms (for example, equating “academic calendar” with “term schedule”) that the keyword system misses, and retrieves more relevant results than keyword-only search. User testing confirmed a marked usability gain: CatSight’s System Usability Scale score averaged 77.17%, versus only 50.33% for the existing system. Overall, these results indicate that CatSight AI outperforms traditional methods in search accuracy and user experience, offering a more accurate, intelligent, and efficient document retrieval solution.

Keywords: *Document Retrieval System, Optical Character Recognition (OCR), Vector-Based Similarity Search, Large Language Model (LLM), Semantic Search*

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Chapter 1 Research Description

In today's era of digital transformation, organizations generate and store an unprecedented volume of documents daily. This exponential growth in document storage has created a pressing demand for efficient document retrieval systems (DRS) to access, manage, and leverage information effectively. A document retrieval system is a specialized application of information retrieval (IR) designed to locate relevant documents within vast repositories based on user queries. By enabling quick and accurate access to information, DRS plays a pivotal role in knowledge management, informed decision-making, and productivity enhancement across diverse domains, including education, business, healthcare, and research (Blair, n.d.; Macdonald & Tait, 2020).

At its core, document retrieval addresses the challenge of identifying stored documents that contain pertinent information. The primary objective is to distinguish between relevant and non-relevant documents for a user's specific query. Modern DRS employs various techniques to achieve this, ranging from traditional keyword-based methods to advanced machine learning approaches. Keyword-based retrieval is a foundational technique that matches user queries to specific keywords within documents. While it is straightforward and easy to implement, its reliance on exact matches can limit its ability to capture the broader context of queries (Blair, n.d.). Boolean retrieval, an early and widely used approach, utilizes logical operators like AND, OR, and NOT to narrow down search results based on exact keyword matches. Though simple, it often lacks the flexibility to handle nuanced user queries (Macdonald & Tait, 2020). The Vector Space Model (VSM) improves on this by representing documents and queries as vectors in a multi-dimensional space, with relevance determined by the cosine similarity between these vectors, enabling more precise ranking of results compared to Boolean methods (Blair, n.d.). Latent Semantic Analysis (LSA) takes this further by uncovering hidden relationships between terms in a document collection through term-document matrix analysis, thereby enhancing retrieval accuracy by capturing the semantic meaning of words rather than relying solely on exact matches (Macdonald & Tait, 2020). Machine learning-based retrieval techniques, such as sup-

port vector machines (SVMs), random forests, and deep learning models, dynamically adapt to user behavior and context, significantly enhancing retrieval accuracy (Blair, 1984). Neural information retrieval systems, leveraging advanced deep learning architectures like recurrent neural networks (RNNs) and transformers, model complex language patterns and contextual nuances in documents, offering even greater precision and relevance in search results (Macdonald & Tait, 2020).

A notable advancement in modern DRS is the integration of Retrieval-Augmented Generation (RAG) frameworks. RAG combines document retrieval with generative AI models to provide highly contextual and human-like responses. This approach retrieves relevant documents from the repository and feeds them into generative models, like GPT, which synthesize coherent and informative answers to user queries. The RAG framework not only enhances the retrieval process but also bridges the gap between static document retrieval and dynamic information synthesis, making it a revolutionary tool in data-driven environments (Macdonald & Tait, 2020).

This research aims to explore the evolving landscape of document retrieval systems, emphasizing their significance in transforming data into actionable knowledge. By analyzing advanced retrieval techniques and highlighting the role of RAG, this study underscores the pivotal role of DRS in fostering innovation and informed decision-making in the digital age.

1.1 Background of the Study

Effective management and access to extensive information repositories are crucial for institutions that maintain vast collections of documents. Document Retrieval Systems (DRS) are essential tools that enable users to efficiently locate specific documents. This study focuses on the Mindanao State University – Iligan Institute of Technology (MSU-IIT), which has its own repositories for Special Orders, Memorandums, and other important records. MSU-IIT maintains two primary repositories:

- **Board of Regents (BOR) Resolutions Archive:** This repository contains BOR resolutions of the MSU System. (Mindanao State University - Iligan Institute of Technology, n.d.-a)
- **IIT Docs:** This repository houses Special Orders, Memorandum Orders, and other

issuances of MSU-IIT. (Mindanao State University - Iligan Institute of Technology, n.d.-b)

Both repositories utilize keyword-based document search mechanisms. IIT Docs, for instance, is built on top of Google Drive and leverages its search capabilities.

While functional, the current keyword-based approach presents several challenges that limit its effectiveness. Keyword searches often fail to account for the context or semantics of user queries, leading to inaccurate or irrelevant search results. Users frequently encounter difficulties in locating exact documents based on their information needs, particularly when dealing with large or complex datasets. This issue is particularly pronounced at MSU-IIT, where reliance on traditional keyword-based models highlights significant gaps in accuracy and efficiency.

In the advent of large language models, the MSU-IIT repository can benefit from enhanced search capabilities that understand the context and semantics of user queries, potentially improving the precision and efficiency of document retrieval.

The current system's gap lies in its inability to comprehend the semantic nuances of user queries, limiting its effectiveness in handling large or complex datasets. As a result, the system is held back in providing a streamlined and user-friendly retrieval process. This inefficiency hinders the institution's ability to fully leverage its vast repository of documents, affecting productivity and decision-making processes.

The study was conducted to address these limitations and explore potential solutions to enhance document retrieval at MSU-IIT. By investigating advanced approaches such as content-based retrieval systems, which will utilize technologies like Optical Character Recognition (OCR), vector embeddings, and Large Language Model (LLM)-powered conversational interfaces, the research will aim to improve search accuracy, user experience, and overall system efficiency. This initiative will seek to bridge the gap between traditional methods and modern technological capabilities, ensuring that MSU-IIT can effectively meet the demands of its users and maintain operational excellence.

1.2 Statement of the Problem

Traditional keyword-based document retrieval systems face global challenges in providing accurate and relevant results due to their lack of contextual and semantic understanding. This often leads to inefficient search experiences for users worldwide. At MSU-IIT, students, faculty, and staff experience similar issues when retrieving institutional documents. The current system's reliance on exact keyword matches limits access to important information. A more advanced, context-aware solution is needed to address these limitations and position MSU-IIT as a model for modern document retrieval.

1.3 Research Objective

The following are the general and specific objects:

1.3.1 General Objective

To develop and evaluate a content-based document retrieval system that integrates Optical Character Recognition (OCR), vector embeddings, and Large Language Model (LLM)-powered conversational interaction.

1.3.2 Specific Objective

1. To collect and annotate representative documents from MSU-IIT's institutional repositories, specifically IIT Docs and BOR Resolutions.
2. To design a content-based document retrieval system architecture integrating OCR for scanned PDFs, vector embeddings for semantic search, and LLMs for conversational queries.
3. To implement the system's key components: OCR processing, semantic search using vector similarity, and an LLM-based chat interface.
4. To evaluate the system's performance in terms of accuracy, user experience, and retrieval speed.
5. To analyze evaluation results to validate system effectiveness and recommend future enhancements.

1.4 Scope and Limitations of the Research

This study focuses on enhancing the Document Retrieval System (DRS) at Mindanao State University - Iligan Institute of Technology (MSU-IIT). The research will involve a sample of 50 PDF documents and will be conducted from January 2025 to April 2025, with participants including MSU-IIT students, faculty, and staff. A notable limitation of the current system is its inability to analyze images and tables within documents, restricting its effectiveness in processing non-textual data. This constraint may impact the retrieval accuracy for documents where critical information is presented in these formats. By acknowledging this limitation, the study aims to provide a clear understanding of the system's current capabilities and identify areas for improvement to effectively meet the institution's document retrieval needs.

1.5 Significance of the Study

This study holds significant value not only for MSU-IIT students, staff, and faculty but also for other institutions and entities that require efficient document storage and retrieval systems. It aims to enhance the overall efficiency and effectiveness of managing and accessing documents within an organization.

Chapter 2 Review of Related Literature

This chapter explores the foundational principles and advancements in document retrieval systems. It examines the traditional title and keyword-based search methods alongside the more sophisticated content-based approaches. A comparative analysis is provided, highlighting the strengths, limitations, and applications of these systems in addressing the evolving demands of users. By delving into the role of OCR, vector embeddings, and LLMs, this review sets the stage for evaluating how content-based systems redefine document retrieval processes in terms of accuracy, efficiency, and user satisfaction.

2.1 Document and Information Retrieval Systems

In the modern era of digital transformation, organizations generate and store vast volumes of documents daily. This exponential growth in document storage has created an urgent need for efficient document retrieval systems (DRS) to access, manage, and utilize information effectively. A document retrieval system is a specialized application of information retrieval (IR) focused on locating relevant documents from a repository based on user queries. It plays a crucial role in knowledge management, decision-making, and enhancing productivity across various domains, including education, business, healthcare, and research.

Document retrieval is the problem of finding stored documents that contain useful information. According to Blaire et al., (n.d.), There exist a set of documents on a range of topics, written by different authors, at different times, and at varying levels of depth, detail, clarity, and precision, and a set of individuals who, at different times and for different reasons, search for recorded information that may be contained in some of the documents in this set. In each instance in which an individual seeks information, he or she will find some documents of the set useful and other documents not useful; the documents found useful are, we say, relevant; the others, not relevant. In order for a person to find all and only the relevant items in a collection of stored documents; one answer is automatic full-text retrieval, which on its surface is disarmingly simple: Store the full text of all documents in the collection on a computer so that every character of every word in every sentence of every document can be located by the machine.

Then, when a person wants information from that stored collection, the computer is instructed to search for all documents containing certain specified words and word combinations, which the user has specified (MacDonald & Tonelleto, 2020).

2.1.1 COIL: Revisit Exact Lexical Match in Information Retrieval with Contextualized Inverted List

In the realm of information retrieval, traditional systems like BM25 have long relied on exact lexical matches, efficiently utilizing inverted list indexes to retrieve documents. However, these systems often struggle with vocabulary mismatches and fail to capture the semantic nuances of language. Conversely, recent neural information retrieval (IR) models emphasize soft semantic matching across all query-document terms, enhancing semantic understanding but at the cost of computational efficiency. Addressing this dichotomy, Gao, Dai, and Callan (2021) introduced the Contextualized Inverted List (COIL) architecture, which aims to integrate the efficiency of exact lexical matching with the semantic depth provided by deep language models.

COIL innovatively stores contextualized token representations within inverted lists, enabling the system to perform exact matches based on the contextual meaning of terms rather than solely on their surface forms. This approach allows COIL to leverage the rich semantic information captured by deep language models while maintaining the retrieval speed characteristic of traditional inverted index structures. By focusing on overlapping query-document tokens' contextualized representations, COIL effectively bridges the gap between lexical and semantic matching. Empirical evaluations demonstrate that COIL outperforms both classical lexical retrievers and state-of-the-art deep language model retrievers, achieving superior retrieval effectiveness with comparable or reduced latency. This performance underscores COIL's capability to deliver precise and contextually relevant results without compromising on efficiency.

The introduction of COIL represents a significant advancement in document and information retrieval systems. By harmonizing the strengths of exact lexical matching and semantic understanding, COIL addresses longstanding challenges such as vocabulary mismatch and semantic ambiguity. Its design offers a practical solution for developing retrieval systems that are both effective and efficient, making it a valuable reference for researchers and practitioners aiming to enhance retrieval performance in various applications.

2.2 Embeddings in Document Retrieval

In the realm of information retrieval, traditional methods have predominantly relied on keyword matching, which often falls short in capturing the nuanced meanings inherent in human language. This limitation has spurred the development of embedding techniques that represent words, sentences, and documents as vectors in high-dimensional space, effectively encoding semantic relationships. By transcending the constraints of exact keyword matches, embeddings facilitate a more profound understanding of context and intent, thereby significantly enhancing the precision and recall of document retrieval systems.

Embeddings represent words, phrases, or documents as vectors in a continuous vector space, enabling systems to measure semantic similarity between textual elements. This representation facilitates the retrieval of documents that are contextually relevant, even when there is no exact keyword match. For instance, embedding-based retrieval systems can recognize that the terms "car" and "automobile" are semantically related, retrieving pertinent documents regardless of the specific term used in the query (Mikolov et al., 2013; Devlin et al., 2019).

Mikolov et al. (2013) introduced word embeddings through the word2vec model, which revolutionized natural language understanding by encoding semantic relationships in vector space. Later advancements, such as BERT (Devlin et al., 2019), extended this concept by producing contextualized embeddings that capture meanings based on surrounding text. These advancements have made embedding-based approaches foundational in modern information retrieval, including models like RepBERT (Zhan et al., 2020), which demonstrated the effectiveness of embeddings in document ranking tasks.

The integration of embeddings into retrieval systems has led to significant improvements in performance. For instance, Huang et al. (2020) implemented embedding-based retrieval in Facebook Search, resulting in enhanced search relevance and user satisfaction. Similarly, Zhan et al. (2020) introduced RepBERT, a model that represents documents and queries with fixed-length contextualized embeddings, achieving state-of-the-art results in initial retrieval tasks.

The adoption of embedding techniques marks a pivotal advancement in document retrieval, enabling systems to move beyond simple keyword matching to a more nuanced understanding of language. This evolution has significantly improved the precision and recall of

retrieval systems, aligning them more closely with human language understanding and enhancing the overall user experience.

2.2.1 Embedding-based Retrieval in Facebook Search

Facebook Search is a feature that enables users to locate content within the Facebook platform, including profiles, pages, groups, posts, and other entities. Unlike traditional web search engines, Facebook Search must consider the unique context of each user, such as their social connections and personal interactions, to deliver relevant results. This personalized approach leverages the user's social graph to enhance the search experience (Huang et al., 2020).

According to Huang et al., (2020), “historically, Facebook Search relied on Boolean matching models, which focus on exact keyword matches. However, this method often failed to capture the nuanced meanings and contextual relevance of queries, leading to less satisfactory user experiences”. To address these limitations, Facebook implemented Embedding-Based Retrieval (EBR) techniques. EBR represents queries and documents as vectors in a continuous vector space, allowing the system to measure semantic similarities between them. This approach enables the retrieval of contextually relevant information, even when there is no exact keyword match.

The adoption of EBR in Facebook Search has significantly improved the relevance of search results and user satisfaction. By understanding the semantic relationships between queries and documents, the system can provide more accurate and personalized results. For instance, if a user searches for “apple,” the system can discern whether the user is interested in the fruit or the technology company based on contextual cues, thereby delivering appropriate results. This advancement aligns with the findings of Huang et al. (2020), who reported significant metric gains in online A/B experiments after implementing EBR in Facebook Search.

In summary, the integration of embedding-based retrieval techniques into Facebook Search has transformed the platform's ability to deliver semantically relevant and personalized search results, thereby enhancing overall user satisfaction.

2.2.2 RepBERT: Contextualized Text Embeddings for First-Stage Retrieval

In the realm of information retrieval, traditional first-stage retrieval methods have predominantly relied on exact term matching techniques, such as bag-of-words models. While these methods are computationally efficient, they often fail to capture the semantic nuances of language, leading to suboptimal retrieval performance. To address these limitations, Zhan et al. (2020) introduced RepBERT, a model that leverages fixed-length contextualized embeddings to represent both queries and documents. By computing the inner products of these embeddings as relevance scores, RepBERT effectively captures semantic relationships, thereby enhancing retrieval accuracy.

According to Zhan, J., et al. (2020), ‘traditional exact term matching methods, such as BM25, depend on the presence of identical terms in both queries and documents. This reliance can result in missed relevant documents that use synonymous or semantically related terms’. RepBERT addresses this issue by encoding the contextual meaning of text into dense vectors, allowing for the recognition of semantic similarities even in the absence of exact term matches. For example, RepBERT can identify that ”car” and ”automobile” refer to the same concept, thus retrieving pertinent documents regardless of the specific term used in the query.

In the study by Zhan, J., Mao, J., Liu, Y., Zhang, M., & Ma, S. (2020), RepBERT represents documents and queries with fixed-length contextualized embeddings, using their inner products as relevance scores. This method achieves state-of-the-art results in initial retrieval tasks, balancing effectiveness and efficiency.

2.2.3 CLEAR

In information retrieval, traditional lexical models like BM25 rely on exact term matching, which can overlook semantic nuances and lead to issues such as vocabulary mismatch. To address these limitations, Gao et al. (2020) introduced the Complement Lexical Retrieval Model with Semantic Residual Embeddings (CLEAR). CLEAR enhances retrieval performance by integrating semantic matching signals from neural embeddings with traditional lexical retrieval methods.

The information regarding the Complement Lexical Retrieval Model with Semantic

Residual Embeddings (CLEAR) is sourced from the study by Gao et al. (2020). In their research, they introduce CLEAR, a retrieval model that enhances traditional lexical retrieval methods like BM25 by incorporating semantic matching signals derived from neural embeddings. This integration is achieved through a novel residual-based embedding learning method, which trains the neural embeddings to capture language structures and semantics that lexical retrieval models may overlook. Empirical evaluations conducted in the study demonstrate that CLEAR outperforms state-of-the-art retrieval models, significantly improving both the accuracy and efficiency of reranking pipelines.

By combining the strengths of lexical and semantic retrieval methods, CLEAR offers a more robust solution to the challenges of information retrieval, effectively bridging the gap between exact term matching and semantic understanding.

2.2.4 Automatic Document Screening of Medical Literature Using Word and Text Embeddings in an Active Learning Setting

In Evidence-Based Medicine (EBM), document screening is essential for providing scientific evidence to support medical decisions. However, the exponential growth of medical literature has made this task increasingly time-consuming and labor-intensive for physicians. To address this challenge, Carvallo et al. (2020) investigated the use of word and text embeddings within an active learning framework to semi-automate the document screening process.

Implementing semi-automated screening methods can significantly alleviate the burden on healthcare professionals, allowing them to focus more on patient care and critical decision-making. By efficiently identifying relevant studies, these methods enhance the quality and timeliness of medical evidence synthesis, ultimately leading to improved patient outcomes. Moreover, the integration of advanced natural language processing techniques, such as word and text embeddings, enables the system to capture complex semantic relationships within the literature, further improving the accuracy of the screening process.

The approach proposed by Carvallo et al. (2020) has practical applications in various medical fields, including the development of clinical guidelines, systematic reviews, and meta-analyses. By reducing the manual effort required for literature screening, it facilitates the timely incorporation of the latest research findings into clinical practice, thereby supporting the con-

tinuous advancement of healthcare quality.

2.3 Information Extraction and Retrieval from PDF Documents

The Portable Document Format (PDF) is a widely adopted standard for disseminating digital documents across various domains, including academia, business, and government. Its design ensures consistent presentation across different platforms and devices, encapsulating text, images, and complex layouts. However, this fixed-layout nature poses challenges for information extraction and retrieval, as the format prioritizes visual fidelity over structural representation.

The significance of information extraction and retrieval from PDF documents lies in its ability to unlock valuable data embedded within these widely used files. By facilitating access to information that might otherwise remain obscured, effective PDF data extraction enhances data accessibility (Livathinos et al., 2021). Transforming unstructured PDF content into structured data further improves search capabilities, allowing users to efficiently locate specific information. This process also supports better knowledge management within organizations, enabling them to manage and utilize extracted information for informed decision-making and streamlined workflows (Clark et al., n.d).

The evolution of information extraction and retrieval from PDF documents reflects significant technological advancements. Initially, information was manually extracted from PDFs, a labor-intensive and error-prone process (Clark et al., n.d.). This was followed by the introduction of rule-based systems, which used predefined rules and templates to identify and extract data. While these systems provided some automation, they struggled with diverse document layouts and lacked scalability. The emergence of machine learning marked a transformative phase, with models capable of learning from annotated data, thereby improving extraction accuracy and adaptability to various document structures (Palm et al., 2017). More recently, deep learning techniques have revolutionized the field by leveraging neural networks to capture complex patterns and relationships within documents. These advancements have significantly enhanced the performance of information extraction systems, enabling more effective processing of PDF content (Gupta et al., 2018).

2.3.1 BEIR

The Benchmarking Information Retrieval (BEIR) dataset is a comprehensive benchmark designed to evaluate the zero-shot performance of information retrieval models across a diverse array of tasks and domains. Introduced by Thakur et al. (2021), BEIR encompasses 18 publicly available datasets, each representing unique retrieval challenges, including fact-checking, question answering, and biomedical information retrieval. This diversity enables researchers to systematically assess how well retrieval models generalize to new, unseen scenarios without domain-specific training.

A key feature of BEIR is its facilitation of zero-shot evaluation, where models are tested on datasets they were not specifically trained on. This approach provides critical insights into a model’s out-of-distribution generalization capabilities, reflecting its robustness and applicability in real-world situations where annotated data may be scarce. By offering a standardized framework, BEIR allows for consistent and fair comparisons among various retrieval systems, including lexical, sparse, dense, late-interaction, and re-ranking architectures.

The significance of BEIR lies in its role as a unifying benchmark that addresses the limitations of previous evaluations, which often focused on narrow or homogeneous settings. By encompassing a wide range of tasks and domains, BEIR challenges models to perform well across different types of retrieval scenarios, thereby promoting the development of more robust and versatile information retrieval systems. The benchmark’s comprehensive nature encourages the advancement of models capable of effective zero-shot retrieval, ultimately contributing to more adaptable and efficient information retrieval technologies.

2.3.2 Declarative Experimentation in Information Retrieval using PyTerrier

In the field of information retrieval (IR), constructing and evaluating complex retrieval pipelines can be a challenging endeavor, often requiring intricate coding and deep system knowledge. To address these challenges, Macdonald and Tonellotto (2020) introduced PyTerrier, a Python-based framework that facilitates declarative experimentation in IR. PyTerrier enables researchers and practitioners to express retrieval pipelines in a manner that closely aligns with their conceptual design, thereby simplifying the experimentation process.

Prior to the development of PyTerrier, the IR community lacked a formalism that allowed for the expressive representation of complex retrieval pipelines in high-level programming languages. Existing tools often required extensive coding and were not conducive to rapid experimentation or reproducibility. Macdonald and Tonellotto identified this gap and proposed PyTerrier as a solution to enable declarative experimentation, thereby addressing the need for a more efficient and collaborative approach to IR research.

The significance of PyTerrier lies in its ability to streamline the development and evaluation of retrieval systems. By allowing users to construct retrieval pipelines declaratively, PyTerrier reduces the complexity associated with traditional procedural programming approaches. This declarative paradigm enhances reproducibility in IR research, as experiments can be easily shared and replicated. Moreover, PyTerrier’s integration with existing IR platforms, such as Anserini and Terrier, enables seamless execution and evaluation of retrieval pipelines, fostering collaboration and innovation within the research community.

2.4 Large Language Models (LLMs) in Information Retrieval

Large Language Models (LLMs) have significantly advanced Information Retrieval (IR), especially in handling complex document formats like PDFs. PDFs are widely used in academia, business, and government because of their consistent visual presentation across platforms, but their fixed layout makes extracting and retrieving information challenging. Earlier methods such as manual extraction, rule-based systems, and early machine learning models often struggled with the diversity and complexity of PDF structures. The introduction of deep learning and neural networks improved this process by allowing systems to recognize complex patterns and better understand document structures.

LLMs like GPT models have further transformed IR by enabling a deeper understanding of natural language and user intent. These models can interpret nuanced queries, retrieve relevant information without relying on exact keyword matches, and support conversational, personalized, and cross-language search. This enhances user experience and makes information more accessible. As LLMs continue to improve, their integration into IR systems promises to make data from documents like PDFs more searchable, accessible, and useful for knowledge management across many fields.

2.4.1 Large Language Models for Information Retrieval: A Survey

Large Language Models (LLMs) have significantly enhanced Information Retrieval (IR) by improving key components such as query rewriting, retrieval, reranking, and reading (Zhu et al., 2023). LLMs excel at interpreting complex user queries by capturing semantic meaning beyond keyword matching, leading to more accurate and relevant document retrieval. They generate contextual embeddings that align with user intent and can rerank documents based on content relevance. Additionally, LLMs synthesize information from multiple sources to provide coherent and informative responses.

The integration of LLMs into IR systems supports natural language interactions, personalized search, multilingual retrieval, and domain-specific applications in fields like healthcare and law (Zhu et al., 2023). However, challenges remain, including the need for large training datasets, limited interpretability, and high computational requirements. Addressing these issues is essential for maximizing the potential of LLMs in improving search relevance and information accessibility.

2.4.2 Fine-Tuning LLaMA for Multi-Stage Text Retrieval

Text retrieval is essential for tasks such as web search, open-domain question answering, and fact verification, and it plays a key role in enhancing large language models (LLMs) within retrieval-augmented generation (RAG) pipelines. This integration helps reduce hallucinations and allows LLMs to access external knowledge (Ma et al., 2024). Recent LLMs like GPT-4 and LLaMA have shown remarkable performance across various NLP tasks, with new methods such as RankGPT, LRL, and PRP using prompting techniques to perform zero-shot listwise or pairwise ranking as generative tasks (Ma et al., 2024).

Ma et al. (2024) explore fine-tuning the LLaMA model for multi-stage text retrieval, which involves an initial coarse retrieval phase followed by refined reranking steps. This approach leverages LLaMA’s advanced language understanding to improve retrieval accuracy, particularly in tasks requiring deep contextual comprehension and precise ranking.

2.5 AI-Powered Prompt-Based Information Extraction

AI-powered prompt-based information extraction enables efficient retrieval of structured data from unstructured text by leveraging Large Language Models (LLMs) with well-crafted prompts (Xu et al., 2023; Çöplü et al., 2024). Unlike rule-based systems, prompt-based methods are flexible and easily adaptable to new tasks by simply modifying the input prompts, reducing development time and effort (Xu et al., 2024; Do et al., 2024). This approach is widely used in fields such as law, healthcare, and business intelligence to extract relevant information from complex documents (Xu et al., 2023; Wei et al., 2023). However, its effectiveness relies heavily on prompt design, as poorly constructed prompts can lead to inaccurate results (Çöplü et al., 2024; Do et al., 2024).

2.5.1 Prompt-Time Symbolic Knowledge Capture with Large Language Models

In their 2024 study, Çöplü et al. address a critical challenge in the deployment of Large Language Models (LLMs) for personalized applications: the models' inherent inability to capture and utilize user-specific knowledge provided during interactions. This limitation hampers the development of AI systems, such as personal assistants, that require the integration of individual user information to function effectively. To tackle this issue, the authors propose a method for prompt-driven symbolic knowledge capture, focusing on the extraction of subject-predicate-object triples from user prompts to construct knowledge graphs.

The researchers explored three approaches to Prompt-to-Triple (P2T) generation to capture symbolic knowledge from prompts effectively. The first approach, Zero-Shot Prompting, leverages the Large Language Model's (LLM) pre-existing knowledge to generate triples without requiring additional training. The second, Few-Shot Prompting, involves providing the model with a limited number of examples to guide the generation process, helping the model better understand the task. Finally, Fine-Tuning adjusts the model's parameters through training on a specialized dataset, significantly improving its ability to extract accurate triples. To evaluate these methods, the authors created a synthetic dataset specifically designed to assess the performance of each approach in capturing symbolic knowledge from user prompts.

The study's experiments reveal that fine-tuning the LLM yields the most accurate and

reliable results in P2T generation, outperforming both zero-shot and few-shot prompting techniques. This indicates that while LLMs possess some capacity for symbolic knowledge extraction through in-context learning, their performance significantly improves when fine-tuned on relevant data.

This research contributes to the field by demonstrating effective strategies for enabling LLMs to capture and utilize user-specific knowledge during interactions. The findings suggest that fine-tuning LLMs can substantially enhance their ability to integrate personal information, thereby improving the functionality of AI systems that depend on personalized data. This advancement is particularly relevant for developing AI applications that require a deep understanding of individual user contexts to provide tailored and accurate responses.

2.5.2 ChatUIE: Exploring Chat-based Unified Information Extraction using Large Language Models

As Large Language Models (LLMs) continue to revolutionize natural language processing (NLP), their ability to perform diverse tasks, including conversational AI and information extraction, has significantly advanced. However, their capacity to extract structured information from complex, domain-specific text remains a challenge, particularly in scenarios involving ambiguous or schema-independent data. To address this gap, Xu et al. (2024) introduced ChatUIE, a unified information extraction framework that builds upon conversational LLM architectures.

In their 2024 study, Xu et al. introduce ChatUIE, a unified information extraction framework built upon ChatGLM, designed to address the limitations of Large Language Models (LLMs) in domain-specific information extraction tasks. While LLMs have demonstrated impressive performance in general conversational contexts, they often struggle with extracting structured information from natural language, especially when it deviates from known schemas or instructions.

ChatUIE employs reinforcement learning to enhance and align various tasks, particularly those involving confusing and limited samples. Additionally, it integrates generation constraints to prevent the inclusion of elements not present in the input, thereby improving the accuracy of the extracted information. Experimental results indicate that ChatUIE significantly

improves the performance of information extraction tasks, with only a slight decrease in general conversational abilities.

This study contributes to the field by providing a framework that effectively combines the conversational strengths of LLMs with enhanced capabilities for domain-specific information extraction. By addressing the challenges associated with extracting structured information from diverse natural language inputs, ChatUIE offers a promising solution for applications requiring precise information extraction across various domains.

2.6 Integration of Embeddings and LLMs for Hybrid Retrieval Systems and Retrieval Augmented Generation Systems (RAGS)

The integration of embeddings and Large Language Models (LLMs) has led to hybrid retrieval systems that enhance information retrieval by combining semantic understanding with contextual generation (Kuzi et al., 2020). Embeddings represent text in vectors to capture semantic similarity, with dense embeddings enabling concept-based matching and sparse embeddings focusing on exact term matches (Sarmah et al., 2024; Yoon et al., 2023). LLMs like GPT-4 contribute contextual comprehension and coherent response generation, making them effective for complex queries (Doan et al., 2024).

Hybrid systems merge dense and sparse retrieval with LLMs to improve search accuracy by balancing semantic depth and keyword precision (Bruch et al., 2022). This approach significantly enhances retrieval performance, especially in handling varied and complex information needs (Kuzi et al., 2020).

2.6.1 Leveraging Semantic and Lexical Matching to Improve the Recall of Document Retrieval Systems: A Hybrid Approach

Kuzi et al. (2020) propose a hybrid retrieval approach that combines semantic matching via deep neural networks with traditional lexical methods. This integration aims to enhance recall by capturing both exact term matches and broader semantic relationships.

In their 2020 study, Kuzi et al. address the limitations of traditional lexical-based retrieval models, such as BM25, which depend on exact keyword matching and may overlook relevant documents that use synonymous or semantically related terms. To overcome these lim-

itations, they propose a hybrid retrieval approach that combines semantic matching, utilizing deep neural networks, with traditional lexical methods. This integration aims to enhance recall by capturing both exact term matches and broader semantic relationships (Kuzi et al., 2020).

The researchers conducted parallel retrieval processes using both semantic and lexical models. The semantic component employed a deep neural network trained to understand complex word relationships, enabling it to identify relevant documents even when there was no direct keyword overlap with the query (Kuzi et al., 2020). The lexical component relied on traditional keyword matching techniques. The results from both models were then merged to form an initial set of documents for re-ranking.

An empirical evaluation using a publicly available TREC collection demonstrated that the semantic retrieval model could retrieve relevant documents not identified by the lexical model. This indicates that the two approaches are complementary, with the hybrid model achieving higher recall than either method alone. The study also found that relevant documents retrieved by the semantic model often exhibited different characteristics compared to those retrieved by the lexical model, underscoring the value of combining both methods.

This study provides valuable insights into the benefits of integrating semantic and lexical matching techniques in document retrieval systems. By demonstrating that a hybrid approach can improve recall and retrieve a more diverse set of relevant documents, it offers a compelling case for the adoption of such models in information retrieval research and practice. The findings suggest that leveraging the strengths of both semantic understanding and exact term matching can lead to more effective retrieval systems, particularly in scenarios where relevant information may be expressed using varied terminology.

2.6.2 A Hybrid Retrieval Approach for Advancing Retrieval-Augmented Generation Systems

In their 2024 study, Doan et al. address the limitations of Retrieval-Augmented Generation (RAG) systems, particularly the challenge of "hallucination"—where models generate responses not grounded in factual data—due to knowledge boundaries and outdated information. They propose a hybrid retrieval method that integrates embeddings from both textual data and knowledge graphs to enhance the retriever component of RAG systems.

The authors' approach involves combining text embeddings with knowledge graph embeddings to capture both the semantic content of passages and the relationships between them. This integration aims to improve the retriever's ability to select relevant information, thereby providing the Large Language Model (LLM) with accurate and contextually appropriate data for response generation. Notably, their method does not require complex joint learning processes, making it more straightforward to implement.

Evaluations on custom test sets demonstrate that this hybrid retrieval approach significantly enhances the accuracy and ranking capabilities of the retriever component (Doan et al., 2024). As a result, the LLM-based reader generates more precise and reliable responses, effectively reducing instances of hallucination. This highlights the potential of the hybrid approach to address critical challenges in RAG systems (Doan et al., 2024).

This study contributes to the field by offering a practical solution to improve the factual accuracy of RAG systems without necessitating extensive fine-tuning or complex training procedures. By leveraging both textual and structured data through knowledge graphs, the proposed method provides a more holistic understanding of the information landscape, which is crucial for applications requiring up-to-date and domain-specific knowledge.

2.6.3 HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction

In their 2024 study, Sarmah et al. introduce HybridRAG, a novel framework that integrates Knowledge Graphs (KGs) with vector-based Retrieval-Augmented Generation (RAG) techniques to enhance information extraction from unstructured financial documents, such as earnings call transcripts. This approach addresses challenges like domain-specific terminology and complex document formats, which often hinder the effectiveness of Large Language Models (LLMs) in financial applications.

HybridRAG combines two complementary retrieval strategies to enhance information extraction. The first, VectorRAG, utilizes vector databases to retrieve semantically relevant textual information, which aids Large Language Models (LLMs) in generating contextually appropriate responses. The second, GraphRAG, employs Knowledge Graphs to capture structured relationships between entities, providing a deeper and more nuanced understanding of the data's

context. By integrating these two approaches, HybridRAG retrieves context from both vector databases and knowledge graphs, enabling LLMs to generate accurate, contextually relevant, and well-informed answers.

Experiments conducted on financial earnings call transcripts, which naturally contain question-answer pairs, demonstrate that HybridRAG outperforms traditional VectorRAG and GraphRAG techniques individually. Evaluations at both the retrieval and generation stages show improvements in retrieval accuracy and the quality of generated answers.

This study highlights the potential of combining unstructured and structured data retrieval methods to enhance the performance of LLMs in complex domains like finance. By leveraging the strengths of both vector-based retrieval and knowledge graphs, HybridRAG offers a more comprehensive approach to information extraction, which can be applied beyond the financial sector to other fields requiring precise and context-aware data retrieval.

2.6.4 Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection

Large Language Models (LLMs) have demonstrated remarkable capabilities across various natural language processing tasks. However, their reliance solely on internal parametric knowledge often leads to factual inaccuracies in generated responses. To address this limitation, Asai et al. (2023) introduced Self-Reflective Retrieval-Augmented Generation (Self-RAG), a framework that enhances LLMs' factual accuracy and overall response quality by integrating on-demand retrieval and self-reflection mechanisms.

Self-RAG operates by training an LLM to perform three key functions: retrieve relevant information, generate responses, and critique its outputs. This process is facilitated through the use of special "reflection tokens" that signal when to initiate retrieval and when to assess the quality of generated content. Upon receiving an input prompt, the model determines the necessity of external information. If additional data is required, it retrieves pertinent passages and incorporates them into the response generation process. Subsequently, the model evaluates its output, identifying and rectifying any factual inconsistencies or errors.

Empirical evaluations across diverse tasks—including open-domain question answering, reasoning, and fact verification—demonstrate that Self-RAG significantly outperforms existing LLMs and retrieval-augmented models. Notably, Self-RAG exhibits superior performance com-

pared to models like ChatGPT and retrieval-augmented Llama2-chat, particularly in enhancing factual accuracy and citation precision in long-form text generation.

The introduction of Self-RAG marks a substantial advancement in the field of retrieval-augmented generation. By enabling LLMs to dynamically retrieve information and critically assess their outputs, Self-RAG effectively mitigates issues related to factual inaccuracies. This approach not only bolsters the reliability of LLM-generated content but also preserves the models' versatility across various applications. The framework's emphasis on self-reflection and adaptive retrieval sets a new precedent for developing more accurate and trustworthy AI language models.

2.6.5 Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity

Large Language Models (LLMs) have significantly advanced natural language processing tasks, particularly in question-answering (QA) systems. However, their reliance on internal parametric knowledge can lead to inaccuracies, especially when handling queries of varying complexities. To address this challenge, Jeong et al. (2024) introduced Adaptive-RAG, a novel framework that dynamically adjusts retrieval-augmented generation strategies based on the complexity of user queries.

The methodology of Adaptive-RAG centers on a classifier—a smaller language model—trained to predict the complexity level of incoming queries. This classifier utilizes automatically collected labels derived from actual model predictions and inherent dataset biases to evaluate query complexity. Based on the assessed complexity, the system dynamically selects the most suitable retrieval strategy. For straightforward queries that can be answered using the LLM's internal knowledge, the system opts for no retrieval, bypassing the need for external data. For moderately complex queries requiring minimal external information, it employs single-step retrieval. For intricate, multi-hop queries necessitating iterative reasoning and extensive data, the system adopts multi-step retrieval. This adaptive approach ensures that each query is processed with optimal efficiency and accuracy, minimizing computational overhead for simple queries while delivering thorough and precise responses for complex ones.

Evaluations on open-domain QA datasets with varying query complexities demonstrated

that Adaptive-RAG enhances both efficiency and accuracy compared to existing baselines, including other adaptive retrieval methods. The framework effectively balances the trade-off between computational resources and response quality, adapting seamlessly to the demands of each query.

Adaptive-RAG represents a significant advancement in retrieval-augmented generation by introducing a dynamic, complexity-aware approach to query processing. By tailoring retrieval strategies to query complexity, it optimizes resource utilization and improves response accuracy. This framework offers a scalable solution adaptable to various QA systems, enhancing their ability to handle a diverse range of user queries effectively.

2.7 LangChain

LangChain is an open-source framework that enhances the development of applications powered by Large Language Models (LLMs), particularly through its support for Retrieval-Augmented Generation (RAG), which combines LLMs with external data retrieval to produce more accurate and context-aware outputs (Wang et al., 2022; LangChain Team, 2022). It provides tools for integrating document loaders, embeddings, and vector stores, enabling the creation of semantic search engines and advanced question-answering systems (LangChain Team, 2022).

LangChain also supports multi-step retrieval pipelines and conversational interactions, allowing developers to build applications that generate reliable and contextually grounded responses (Wang et al., 2022; LangChain Team, 2022). Overall, it serves as a powerful bridge between LLMs and retrieval systems, improving the performance and applicability of information retrieval solutions across various domains.

2.7.1 Automating Customer Service with LangChain: A Custom Open-Source GPT Chatbot

In the era of digital transformation, organizations increasingly rely on advanced technologies to enhance customer service efficiency and user satisfaction. Traditional methods, such as static Frequently Asked Questions (FAQs), often fail to meet the dynamic and diverse needs of users. To address these limitations, Pandya and Holia (2023) proposed Sahaay, a custom open-source

chatbot framework powered by LangChain and Large Language Models (LLMs). Sahaay aims to revolutionize customer service by providing real-time, personalized, and context-aware support.

The development of Sahaay involves several key components. First, data collection is achieved through web scraping techniques to gather comprehensive and up-to-date information from organizational websites, ensuring the chatbot has a robust knowledge base. Second, embeddings and language models play a pivotal role, with textual data represented using embeddings and knowledge retrieval facilitated by Google's Flan T5 models (XXL, Base, and Small). These models enable Sahaay to generate accurate and contextually relevant responses. Finally, integration with customer service platforms ensures seamless embedding of the chatbot into existing support infrastructures, allowing for real-time query resolution and enhanced user interaction. This multi-faceted approach equips Sahaay to handle diverse queries effectively while aligning with organizational needs.

According to Pandya, S., & Holia, R. (2023), the implementation of Sahaay within an educational institution demonstrated its capability to handle diverse queries from prospective and current students, as well as researchers. The chatbot effectively provided information on topics such as notice board updates and potential research guides, showcasing its versatility and responsiveness.

This study contributes to the field of document and information retrieval systems by illustrating how integrating LLMs with open-source tools like LangChain can revolutionize customer service. By moving beyond static FAQs to dynamic, personalized interactions, organizations can enhance customer satisfaction and engagement. The research underscores the potential for scalable solutions across various industries, paving the way for more responsive and efficient customer service ecosystems.

2.7.2 Generating Breast Ultrasound Reports Using LangChain for Standardized Medical Documentation

Medical imaging plays a critical role in the early detection and diagnosis of various conditions, including breast cancer. Breast ultrasound (BUS), as a non-invasive imaging technique, is widely used for its ability to identify abnormalities in breast tissue. However, generating de-

tailed and standardized medical reports from BUS images is a labor-intensive process that places significant demands on radiologists. To address these challenges, Huh, Park, and Ye (2023) propose an innovative system leveraging LangChain and Large Language Models (LLMs) to automate the generation of BUS reports. Their approach focuses on enhancing efficiency, consistency, and quality in medical documentation by integrating multiple image analysis tools within the LangChain framework.

The proposed system employs a combination of designated tools and LLM-powered text generation within LangChain to extract relevant features from BUS images. These features are then contextualized within a clinical framework to produce comprehensive and standardized medical reports (Huh et al., 2023). According to Huh et al. (2023), by automating the traditionally manual process of feature extraction and interpretation, this approach reduces the workload on radiologists while ensuring uniformity in the generated reports.

Extensive experiments conducted by the authors demonstrate that the integrated tools deliver significant improvements in both qualitative and quantitative metrics. Clinical evaluations confirm the clinical utility of the generated reports, highlighting their alignment with diagnostic needs and their potential for practical application in real-world healthcare settings (Huh et al., 2023).

This study is a significant contribution to the field of medical imaging and documentation. By leveraging LangChain and LLMs, the authors present a scalable solution that integrates multiple image analysis tools for automated report generation. This advancement not only enhances the efficiency of diagnostic workflows but also ensures standardized documentation, which is critical for improving patient care outcomes (Huh et al., 2023). Furthermore, the approach outlined in this study demonstrates the adaptability of LangChain for other medical imaging modalities, providing a foundation for further innovations in healthcare technology.

2.7.3 Chatbot Technology in Higher Education: Leveraging LangChain and GPT for Internationalization and Digital Transformation

The internationalization of higher education institutions (HEIs) necessitates innovative approaches to support a diverse student body effectively. Traditional student support services often face challenges in scalability and personalization, particularly for international students

who may encounter language barriers and cultural differences. In response to these challenges, Hsain and El Housni (2024) explore the integration of advanced chatbot technology powered by GPT-3.5 and GPT-4 Turbo to enhance student engagement and information access. Their research focuses on leveraging Large Language Models (LLMs) to facilitate digital transformation in higher education, aiming to provide real-time, personalized support that aligns with the needs of a global student population.

The study employs a comprehensive technological stack, including Python 3, GPT API, LangChain, and Chroma Vector Store, to develop a chatbot capable of delivering context-aware and accurate responses. A critical component of the development process is the creation of a high-quality, timely, and relevant transcript dataset, which serves as the foundation for training and testing the chatbot's performance. This dataset encompasses a wide range of topics pertinent to international students, ensuring that the chatbot can address diverse queries effectively.

Evaluation of the chatbot's performance indicates a high level of efficacy in providing comprehensive and relevant responses. User feedback suggests a preference for the chatbot over traditional support methods, citing its ability to engage in real-time interactions and maintain conversational memory, which enhances the overall user experience. Additionally, the chatbot demonstrates a low error rate, further validating its reliability as a tool for student support.

This study contributes to the field of educational technology by demonstrating the potential of LLM-powered chatbots to transform student support services in HEIs. By facilitating real-time, personalized interactions, such chatbots can significantly improve accessibility and satisfaction among international students. The research underscores the importance of digital transformation in higher education and provides a framework for integrating advanced AI technologies to meet the evolving needs of a diverse student body.

2.7.4 A Conversational Agent for Promoting Cultural Awareness in Seoul Using LangChain

Promoting cultural awareness among visitors is essential for preserving and appreciating a city's heritage. In Seoul, a city rich in historical sites, effectively disseminating information about these landmarks can enhance tourists' experiences and foster a deeper understanding of Korean culture. To address this need, Suh, Kwak, Kim, and Cho (2024) developed a conversational

agent designed to provide accessible and accurate information about Seoul’s historical sites. This agent leverages LangChain, a framework for building applications powered by large language models (LLMs), to deliver contextually relevant responses to user inquiries.

The development of the conversational agent involved several key components that ensured its functionality and user engagement. First, data collection was conducted using information provided by the Seoul Metropolitan Government, guaranteeing the dataset was authoritative, up-to-date, and rich with details about various historical sites, their significance, and their locations. Next, framework implementation utilized LangChain to integrate multiple language models and tools, enabling the agent to process and generate natural language responses effectively. This framework facilitated the seamless combination of different models, enhancing the agent’s conversational capabilities. Finally, user interface development was achieved using Streamlit, an open-source app framework that provided an interactive and user-friendly platform. This design allowed users to engage effortlessly with the agent, receiving accurate and contextually relevant information conversationally.

Despite the limited volume of data, the conversational agent consistently delivered reliable and accurate responses aligned with the available information. The system effectively increased awareness among visitors unfamiliar with Seoul’s cultural heritage, providing precise locations and historical context for various sites. The use of LangChain enabled the agent to handle diverse queries, demonstrating its versatility in managing complex conversational scenarios.

This study highlights the potential of integrating LLM frameworks like LangChain with conversational agents to promote cultural heritage. By providing accessible information through natural language interactions, such systems can enhance tourists’ experiences and contribute to cultural preservation. The research also underscores the importance of utilizing authoritative data sources and user-friendly interfaces to ensure the effectiveness of conversational agents in real-world applications.

2.8 Challenges and Limitations

Despite advancements, challenges persist in document retrieval systems, including handling ambiguous queries, ensuring scalability, and maintaining the interpretability of complex models.

Balancing precision and recall remains a critical concern, especially when integrating neural embeddings and LLMs

2.9 Future Trends in AI-Powered Document Retrieval

Future trends point towards deeper integration of LLMs and embeddings, development of more robust hybrid models, and enhanced capabilities in zero-shot and few-shot learning scenarios. Emphasis on ethical considerations and transparency in AI-powered retrieval systems is also expected to grow.

2.10 Summary of Related Literature

Table 2.1. *Summary Features of Related Literature*

Study	DR (2.1)	EMB (2.2)	PDF (2.3)	LLM (2.4)	PIE (2.5)	HYB (2.6)	LC (2.7)
Gao, L., Dai, Z., & Callan, J. (2021)	✓	✓					
Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019)	✓	✓		✓			
Huang, P.-S., et al. (2020)	✓	✓				✓	
Zhan, J., Mao, J., Liu, Y., Zhang, M., & Ma, S. (2020)	✓	✓					
Carvallo, A., Parra, D., Lobel, H., & Soto, A. (2020)		✓	✓				
Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., & Gurevych, I. (2021)	✓	✓					
Macdonald, C., & Tonellotto, N. (2020)	✓						
Feng, J., Tao, C., et al. (2024)	✓			✓	✓	✓	
Çöplü, T., et al. (2024)			✓	✓	✓		
Xu, J., et al. (2024)	✓	✓	✓	✓	✓		✓
Doan, N. N., et al. (2024)	✓	✓		✓	✓	✓	✓
Kuzi, S., et al. (2020)	✓	✓		✓		✓	
Sarmah, B., et al. (2024)	✓	✓		✓		✓	
Asai, A., et al. (2023)	✓			✓		✓	
Jeong, S., et al. (2024)	✓	✓		✓		✓	
Wang, J., et al. (2022)	✓	✓		✓	✓	✓	✓
Pandya, S., & Holia, R. (2023)				✓	✓		✓
Huh, S., Kim, J., & Park, S. H. (2023)			✓	✓	✓		✓
Hsain, F., & El Housni, Z. (2024)				✓	✓		✓
Suh, J., Lee, Y., & Park, M. (2024)				✓	✓		✓
Proposed System	✓	✓	✓	✓	✓	✓	✓

Note. DR = document and information retrieval systems; EMB = embeddings in document retrieval; PDF = information extraction and retrieval from PDF documents; LLM = large language models in information retrieval; PIE = AI-powered prompt-based information extraction; HYB = integration of embeddings and LLMs for hybrid retrieval and RAG; LC = LangChain.

Chapter 3 Theoretical Framework

This chapter lays the conceptual foundation for the development of CatSight AI. The study integrates theories and models from the domains of information retrieval (IR), human-computer interaction (HCI), and large language models (LLMs), ensuring the system addresses both technical effectiveness and user satisfaction.

3.1 Information Retrieval Theory

Information Retrieval (IR) Theory is the study of methods and systems for finding relevant information from large collections of data, especially unstructured data such as text. It forms the foundation of technologies like search engines, digital libraries, and question-answering systems. At its core, IR involves documents—units of data stored in the system—and queries, which represent users’ information needs. To enable efficient retrieval, documents undergo preprocessing through indexing, which includes tokenization, stopword removal, stemming or lemmatization, and the construction of inverted indexes.

3.1.1 Tokenization

Tokenization in Information Retrieval (IR) is the process of breaking down text into smaller units called tokens, typically words or meaningful terms, which serve as the basic building blocks for indexing and matching documents to queries. It is a crucial preprocessing step that standardizes text by removing punctuation, handling case sensitivity, and separating compound words, thereby enabling efficient and accurate information retrieval. For instance, the sentence “Information retrieval is powerful” would be tokenized into [“Information”, “retrieval”, “is”, “powerful”], allowing the IR system to analyze and compare textual content effectively. The complexity of tokenization can vary depending on the language—English tokenization is relatively straightforward, whereas languages like Chinese or Japanese require more sophisticated segmentation techniques. Tokenization facilitates the construction of structures like inverted indexes and significantly impacts retrieval performance. For a comprehensive discussion of tokenization, Jurafsky and Martin (2023).

3.1.2 Cosine Similarity in Semantic Search

Cosine similarity is a method used in semantic search to measure how similar two texts are by comparing their vector representations. In CatSight AI, it helps match the meaning of user queries with institutional documents using vectors generated from language models. This method focuses on the angle between vectors, not their length, allowing for more accurate, context-aware search results than traditional keyword matching (Huang, 2008). **Equation 1**, shows the equation of getting the cosine similarity of the semantic search.

$$\begin{equation}$$

$$\text{Cosine Similarity} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \times |\vec{B}|}$$

$$\end{equation}$$

Explanation:

$\vec{A}$ and $\vec{B}$ denote vectors A and B

$\cdot$ represents the dot product

$|\vec{A}|$ and $|\vec{B}|$ are the magnitudes (norms) of the vectors

3.2 Large Language Models in IR

Recent advancements in transformer-based LLMs such as BERT (Devlin et al., 2018) and GPT (Radford et al., 2019) have revolutionized retrieval-based tasks through semantic search and contextual query understanding. CatSight AI adopts such LLMs to handle open-ended queries with improved precision, echoing approaches validated in contemporary research (Gao et al., 2021).

3.2.1 LLM Hallucinations

Hallucination in Large Language Models (LLMs) refers to the generation of text that is fluent and contextually relevant but factually incorrect or fabricated. This phenomenon poses a significant challenge in systems like CatSight AI, which rely on LLMs to interpret and retrieve information from documents. Since LLMs are trained on vast, mixed-quality datasets using statistical language modeling, they generate responses based on learned word patterns rather than

verified facts. Without grounding in external knowledge bases or the original document context, LLMs may produce answers that appear relevant but do not accurately reflect the source material. This issue is particularly critical in document retrieval systems, where factual accuracy and source traceability are essential. Furthermore, reinforcement techniques like RLHF (Reinforcement Learning from Human Feedback) may unintentionally favor responses that are convincing over those that are correct. Understanding and mitigating hallucination is thus crucial to improving the reliability and trustworthiness of CatSight AI’s LLM-based chat interface (Ji et al., 2023).

3.3 System Evaluation

3.4 Cognitive Load Theory and Usability

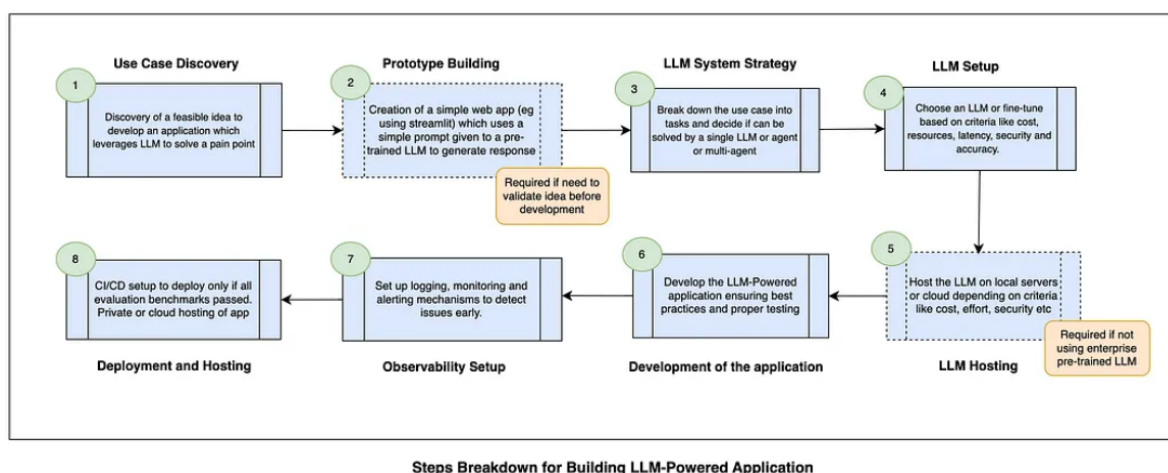
Drawing from Cognitive Load Theory (Sweller, 1988), CatSight AI is designed to minimize user confusion through a clean interface and structured interactions. This theory guides UI/UX choices, ensuring users can process and retrieve information without cognitive overload (Paas & Van Merriënboer, 1994).

3.5 Summary of Theoretical Model

The integrated framework—composed of IR principles, LLM-driven search, TAM for adoption, and cognitive load for usability—guides both the functional and experiential design of CatSight AI. This synthesis ensures a technically sound, user-friendly system tailored to MSU-IIT’s academic needs.

Chapter 4 Research Methodology

This chapter outlines the methodology used to develop CatSight AI, a document retrieval system (DRS) tailored for Mindanao State University – Iligan Institute of Technology (MSU-IIT). Adapted from Pallavi Sinha’s framework for building LLM-powered applications, the process excludes deployment-related steps to align with the project’s timeline (January–April 2025) as shown in **Figure 2**. However, this study only follows six key phases. The structured approach ensures that CatSight AI meets MSU-IIT’s specific needs, following best practices in AI development such as iterative testing and data-driven refinement. Only six key phases were



Steps Breakdown for Building LLM-Powered Application

Figure 4.1. *Steps Breakdown for Building LLM-Powered Applications (Sinha, 2024)*

adopted to ensure the feasibility and relevance of the system during its limited development window. This structured approach ensures that CatSight AI meets MSU-IIT’s specific needs, following best practices in AI development, such as iterative testing, ethical considerations, and data-driven refinement (Amershi et al., 2019).

4.1 Use Case Discovery

The proponents observed that MSU-IIT’s current document retrieval system posed challenges for students, faculty, and staff. The keyword-based search often failed to return relevant results when queries didn’t match exact terms, making it difficult to find necessary documents. For instance, searching for “policy updates” might miss files labeled differently, such as “revised regulations.” These limitations revealed the system’s lack of contextual understanding

and highlighted the need for a more intelligent solution. CatSight AI addresses this by integrating OCR, vector similarity search, and LLM-based chat interfaces, enabling semantic search and conversational interaction to improve search accuracy and user experience.

4.1.1 Process of Document Collection and Preparation

Building on the identified limitations of MSU-IIT’s traditional document retrieval system, the study moves toward developing and evaluating a more intelligent solution—CatSight AI. To ensure that the system is tested under realistic and representative conditions, the process begins with the careful collection and preparation of institutional documents. These documents are sourced from two main repositories: (1) MSU-IIT’s internal document archives and (2) the MSU System Board of Regents (BOR) Resolutions.

Both scanned and text-based PDF files are gathered, curated, and pre-processed to serve as the dataset for evaluating the proposed Document Retrieval System (DRS). This step ensures the inclusion of a wide variety of document types, formats, and complexities, particularly those that require Optical Character Recognition (OCR) for semantic interpretation.

To maintain relevance and comprehensiveness, specific criteria were observed in selecting the documents:

- **Relevance:** Files were chosen based on their alignment with commonly referenced administrative functions and institutional topics.
- **Diversity of Content:** The dataset includes various categories such as policies, appointments, and event-related issuances.
- **Format Variety:** Both scanned and digitally native PDFs were included to assess the system’s performance in handling OCR and different file structures.
- **Structural Complexity:** Documents featuring tables, multiple sections, and complex layouts were prioritized to rigorously test the DRS’s retrieval capabilities.

This comprehensive document preparation process sets the foundation for a meaningful evaluation of CatSight AI’s ability to overcome the limitations of the current system and deliver more accurate, context-aware search results.

4.1.2 Problem Definition & Stakeholders

The development of CatSight AI was motivated by critical limitations observed in the existing Document Retrieval System (DRS) at Mindanao State University - Iligan Institute of Technology (MSU-IIT). The institution relies on two primary repositories: the Board of Regents (BOR) Resolutions Archive, which governs MSU System policies, and IIT Docs, a repository of Special Orders, Memorandum Orders, and administrative issuances built on Google Drive's keyword-based search. While functional, these systems exhibit significant shortcomings:

1. Semantic Blindness: The keyword-only mechanism fails to interpret contextual nuances or semantic relationships, leading to irrelevant or incomplete results. For example:
 - Queries for "Special Orders" miss documents labeled with abbreviations like "SO."
 - Synonym-rich content (e.g., "academic calendar" vs. "term schedule") remains undiscovered.
 - Paraphrased queries (e.g., "student allowances" vs. "financial aid") yield inconsistent results.
1. Format Exclusion: Scanned or image-based PDFs are entirely excluded from search results due to the lack of Optical Character Recognition (OCR) capabilities. Documents with complex layouts—such as tables, multi-column text, or handwritten annotations—are similarly unsearchable.
2. Rigid Interaction: Users cannot refine queries iteratively due to the absence of a conversational interface, forcing them to rely on repetitive keyword adjustments.

Primary Stakeholders:

- Students: Require timely access to academic policies (e.g., scholarship resolutions, enrollment guidelines) and event announcements.
- Staff: Responsible for locating memorandums (e.g., procurement procedures, compliance records) to streamline operational workflows.

4.2 Agent Prototype Building

4.3 LLM System Strategy

Strategic design decisions were made, including model selection, embedding strategies, and chunking techniques for documents. The LLM System Strategy for CatSight AI prioritized a single LLM approach to streamline development and reduce computational overhead, opting against multi-agent systems that are better suited for complex, specialized tasks (Chen et al., 2023). This decision was informed by a rigorous evaluation of multiple candidate models from **Ollama** to ensure optimal performance for document retrieval and conversational interaction at MSU-IIT. Several models were tested for both the primary LLM and the embedding model:

For the LLM:

- **llama3.1:8b**: Demonstrated strong language understanding but required significant computational resources, making it less feasible for real-time applications.
- **llama3.2:1b**: A lightweight model with low resource demands but insufficient for handling complex conversational tasks requiring deep contextual understanding.
- **qwen3:1.7b**: Offered good language capabilities but struggled with processing long-context documents, a critical requirement for MSU-IIT's use case.
- **qwen2.5:7b-instruct-q4_K_M**: Excelled in instruction-following tasks but was less efficient for real-time interaction due to higher latency.
- **llama3.1:8b-text-fp16**: Optimized for text processing, it performed well but was less versatile than Llama 3.1 for conversational applications but requires more resources.
- **llama3.18b-q4**: Selected as the primary LLM for its balanced performance, efficiency, and ability to handle complex queries and generate context-aware responses, fine-tuned for MSU-IIT's institutional context (Meta AI, 2025).

For the embedding model:

- **bge-m3:latest**: Initially evaluated for its multi-lingual support but found lacking in accurately capturing specific entities such as names, numbers, and native terms prevalent in MSU-IIT's documents.
- **mxbai-embed-large**: Chosen for its superior semantic accuracy, particularly in handling nuanced entities, making it ideal for semantic search across MSU-IIT's diverse

document set (mixedbread.ai, 2024).

Benchmarking was conducted using sample documents from MSU-IIT, focusing on tasks such as text generation, question answering, and semantic similarity search. Metrics included response accuracy, latency, and resource utilization. Llama 3.1 8B Q4 outperformed other LLMs in generating accurate and contextually relevant responses with moderate computational requirements, making it suitable for real-time applications. Similarly, mxbai-embed-large demonstrated higher precision in embedding tasks compared to bge-m3, particularly for documents containing names, numbers, and native terms, ensuring robust retrieval performance (Brown et al., 2020; mixedbread.ai, 2024).

4.4 LLM Setup

One of the key processes of this study is choosing the right LLM setup. This section discusses the LLM setup chosen by the proponents:

4.4.1 LLM Underlying Process

According to Mulugeta (2023), a Large Language Model (LLM) operates through a series of structured processes to interpret and generate human-like text. A simplified representation of this workflow is illustrated in the following image

IMAGE

The article then provides a deeper exploration of several key steps in the process while intentionally excluding more advanced topics—such as model training—that fall outside the scope of the current discussion.

4.4.2 Ollama

Ollama is an open-source framework optimized for local deployment and efficient inference of large language models (LLMs). In CatSight AI, Ollama serves as the foundational LLM engine, running entirely on-premises within MSU-IIT's infrastructure to guarantee compliance with the Philippine Data Privacy Act. It integrates seamlessly with Django's REST framework to expose inference endpoints and with our Retrieval-Augmented Generation (RAG) pipeline for context-aware academic document retrieval. This setup balances low-latency performance, domain-specific customization (e.g., adherence to MSU-IIT's document schemas and terminologies),

and institutional governance requirements—ensuring a scalable, privacy-preserving AI solution tailored to an academic environment.

4.4.3 Llama 3.1:8b-Q4

Llama 3.1’s 8-billion-parameter variant, quantized to 4 bits per weight (Q4), offers nearly FP16-level performance (□99.3 % relative quality) while reducing memory footprint by over 4×—making on-premises inference feasible on modest hardware of 4–8 GB RAM (Meta AI, 2024). This quantized model runs natively in Ollama, enabling CatSight AI to deliver fast, context-aware responses without offloading data externally.

4.4.4 mxbai-embed-large

The mxbai-embed-large model from Mixedbread AI produces dense, 768-dimensional sentence embeddings optimized for semantic retrieval. Trained without overlap on the MTEB benchmark, it outperforms many same-sized and even larger proprietary models (Mixedbread AI, 2024). This ensures high-quality vector representations for MSU-IIT’s diverse document corpus. In CatSight AI, these embeddings power our pgvector store, underpinning precise similarity search in the RAG pipeline.

4.5 LLM Hosting

CatSight AI’s core LLM is hosted on-premises via Ollama, ensuring all document embeddings and inference remain within MSU-IIT’s infrastructure to uphold institutional data-privacy requirements (Philippine Data Privacy Act compliance). Ollama’s lightweight local runtime provides cloud-like elasticity—seamlessly scaling to accommodate peak query loads, while exposing inference endpoints through Django’s REST framework. This integration delivers low-latency, auditable API calls for document retrieval, fits naturally into CatSight AI’s unified agent architecture, and aligns with MSU-IIT’s governance policies around data sovereignty.

4.6 Development of the System

This section outlines the technical implementation of the system. Under this section are the dependency management, system algorithm, system functions, environment, and resources that underpin the system operation.

4.6.1 Dependency Management and Setting up Development Environment

CatSight AI leverages **Docker** to package the application and all its dependencies into a containerized environment. Docker is an open platform for developing, shipping, and running applications in lightweight, self-contained units called containers. By defining a **Dockerfile** that lists the base image, required system packages, Python runtime, and Python libraries (in a ‘requirements.txt’), every developer and deployment environment uses the exact same setup, eliminating “it works on my machine” issues. Containers isolate CatSight AI’s runtime from host-level variations, streamline onboarding (one ‘docker build’ + ‘docker run’ command), and support consistent CI/CD pipelines across development, testing, and production stages (Docker, 2024).

4.6.2 System Algorithm

The core logic of the system is captured in three complementary algorithms: the Document Process Algorithm, which the user uploads files which will then be converted into Markdown, then from Markdown, it will be broken down using *RecursiveTextSplitter* from *Langchain* for context-aware retrieval, then the retrieved markdown will be embedded, then the contents that have been embedded will be summarized which will also result to metadata extraction, the Search-Retrieval Algorithm, which handles both traditional keyword and semantic (AI-powered) document searches, and the Interactive-Chat Algorithm, which enables users to engage in dynamic, conversational queries for more contextual and human-like interactions. Each user request includes the input query along with session-specific metadata, enabling personalized and efficient response generation.

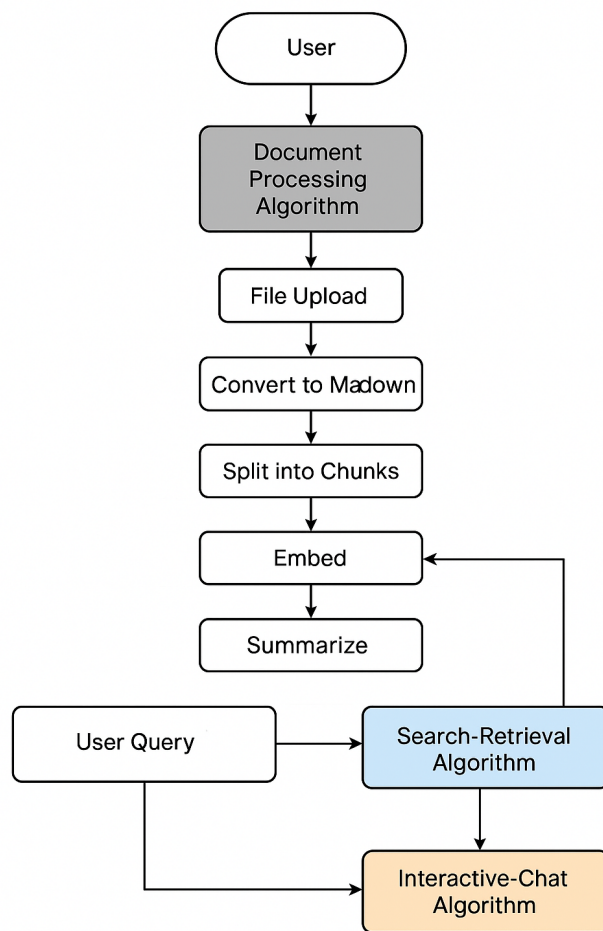
4.6.2.1 Document Process Algorithm

4.6.2.2 Search-Retrieval Algorithm

4.6.2.3 Interactive-Chat Algorithm

4.6.3 Flowchart of CatSightAI

The chatbot engine processes user input and generates responses through a structured, LLM-powered workflow, as illustrated in Fig. 3.17. Each user message follows these stages:



System Workflow

Figure 4.2. *System Workflow*

4.6.4 System Workflow Diagram

4.6.5 System Functions

4.6.6 Environment and Resources

Chapter 5 Design and Implementation Issue

This chapter discusses the design and implementation of the major data structures and algorithms used in the software. It includes a discussion on the major issues and problems encountered, and the corresponding solutions and alternatives employed by the proponents. Parts of the design tools in the Technical Manual may be lifted as figures in this chapter.

This chapter discusses the design and implementation of the

Chapter 6 Results and Observations

Chapter 7 Conclusion and Recommendations

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Appendix B Personal Vitae

Arellano, Carlo P.

[TODO: Vitae from the defended manuscript (Appendix B): photo, contact, education, skills.]

Carnaje, Michael James M.

[TODO: Vitae from the defended manuscript (Appendix B): photo, contact, education, skills.]

Lavesores, Fulgent Kvasir E.

[TODO: Vitae from the defended manuscript (Appendix B): photo, contact, education, skills.]